

Let, x be an arbitrary element of $A \cap \bar{B}$, i.e. $x \in A \cap \bar{B}$.

By Defn. 2, $x \in A \wedge x \in \bar{B}$. By Defn. 5, $x \notin A \wedge x \notin B$. By Defn. of not belongs to, $\neg(x \in A) \wedge \neg(x \in B)$. By 2nd De Morgan's Law of Propositions, $\neg(x \in A \vee x \in B)$. By Defn. 1, $\neg(x \in A \cup B)$. By defn. of not belongs to, $x \notin A \cup B$. By Defn. 5, $x \in \overline{A \cup B}$. By Defn. 3 of Sec-2.1, $\overline{A \cup B} \subseteq \overline{A \cap B}$. $\therefore A \cup B = \overline{\overline{A \cup B}} = \overline{\overline{A \cap B}} = A \cap B$ $\square$

b)

A	B	$A \cup B$	$\overline{A \cup B}$	$\bar{A}$	$\bar{B}$	$\overline{A \cap B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

16) Let A and B be sets. Show that

a) $(A \cap B) \subseteq A$.

Soln: Let, x be an arbitrary element of $A \cap B$, i.e. $x \in A \cap B$.
By Defn. 2, $x \in A \wedge x \in B$. By defn. of AND, $x \in A$ and $x \in B$.
By Defn. 3 of Section 2.1, $(A \cap B) \subseteq A$ $\square$

b) $A \subseteq (A \cup B)$

Soln: Let, x be an arbitrary element of A , i.e. $x \in A$.
By Defn. of OR, $x \in A \vee x \in B$. By Defn. 1, $x \in A \cup B$.
By Defn. 3 of Section 2.1, $A \subseteq A \cup B$ $\square$

c) $A - B \subseteq A$

Soln: Let, x be an arbitrary element of $A - B$, i.e. $x \in A - B$.
By Defn. 4, $x \in A \wedge x \notin B$. By defn. of AND, $x \in A$ and $x \notin B$.
By Defn. 3 of Section 2.1, $A - B \subseteq A$ $\square$

d) $A \cap (B - A) = \emptyset$

Soln: Let, x be an arbitrary element of $A \cap (B - A)$, i.e. $x \in A \cap (B - A)$.
By Defn. 2, $x \in A \wedge x \in B - A$. By Defn. of difference, $x \in A \wedge (x \in B \wedge x \notin A)$.
By 2nd Commutative Law of propositions, $x \in A \wedge (x \notin A \wedge x \in B)$.